

# Thematic maps with mapsf : : CHEAT SHEET



Create and integrate thematic maps in your workflow.

## Base map

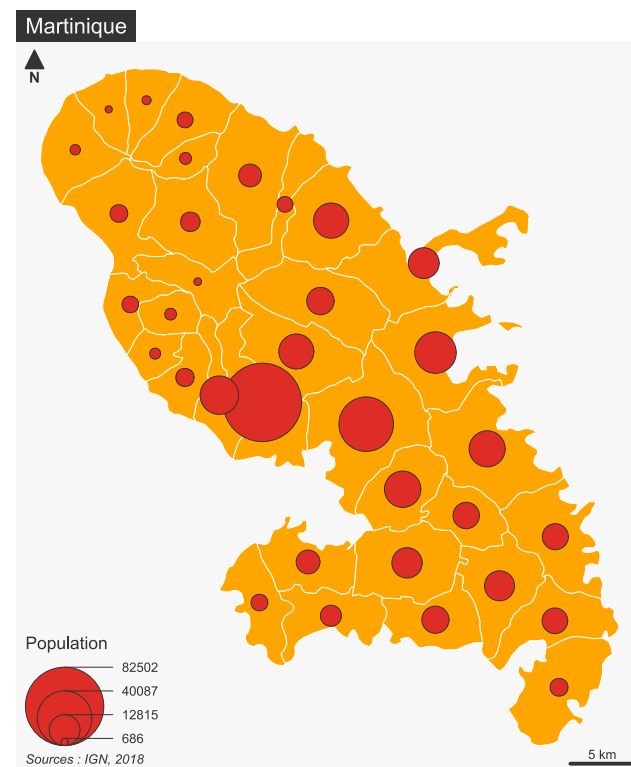
Import library  
`library(mapsf)`

Import the sample data set  
`mtq <- mf_get_mtg()`

Initiate a base map centered on a specific extent  
`mf_map(x = mtq, col = "orange",  
border = "white")`

Plot symbology  
`mf_map(x = mtq, type = "prop", var = "POP",  
leg_title = "Population", add = TRUE)`

Complete layout (credits, title, north, arrow, scale bar)  
`mf_layout(title = "Martinique",  
credits = "Sources: IGN, 2018")`



## Colors

mapsf can use color palettes from `hcl.colors()`.  
`mf_get_pal()` is useful to create well-balanced asymmetric diverging palettes



## Symbology

The x argument should be an sf object.  
Input geometries can be polygons, lines or points.

 Choropleth (ratios)  
`mf_map(x = mtq, type = "choro",  
var = "var", method = "quantile")`

 Typology (categories)  
`mf_map(x = mtq, type = "typo", var = "var")`

 Proportional Symbols (stocks)  
`mf_map(x = mtq, type = "prop", var = "var",  
inches = 0.1, symbol = "circle")`

 Graduated Symbols (stocks)  
`mf_map(x = mtq, type = "grad", var = "var",  
pch = 24)`

 Symbols (categories)  
`mf_map(x = mtq, type = "symb", var = "var",  
pch = c(21:23))`

 Choropleth proportional symbols (stocks - ratios)  
`mf_map(x = mtq, type = "prop_choro",  
var = c("var1", "var2"))`

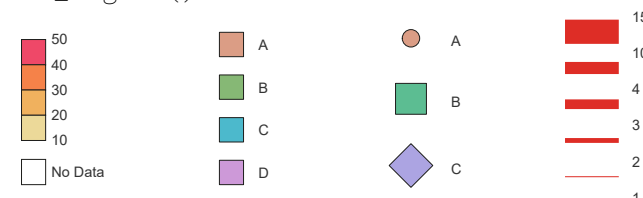
 Colorized proportional symbols (stocks - categ.)  
`mf_map(x = mtq, type = "prop_typo",  
var = c("var1", "var2"))`

 Choropleth symbols (ratios - categories)  
`mf_map(x = mtq, type = "symb_choro",  
var = c("var1", "var2"),  
pch = c(21:23))`

 Raster  
`mf_raster(x = raster)`

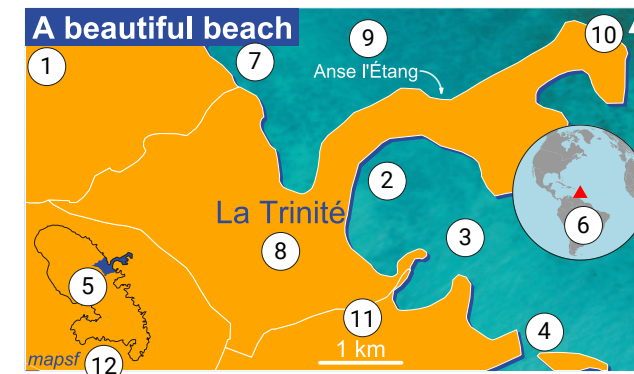
## Legends

Default legends are plotted along maps.  
Customization parameters are available with  
`mf_legend()`



## Map Layout

Along with cartographic functions, other functions are dedicated to customize the layout design.



① Set a map theme (figure margins, colors, title options...)  
`mf_theme(bg = "white", tab = TRUE,  
mar = c(0,0,0,0), pos = "left")`  
Builtin themes are available : default, ink, dark, agolalight, candy, darkula, iceberg, green, nevermind, jsk or barcelona.

② Init a map centered on a specific area  
`mf_init(x = mtq[30, ])  
mf_map(x = mtq, col = "orange",  
border = "white", add = TRUE)`

③ Import external image for background  
`mf_background(filename = "img/sea.jpg")`

④ Create a shadow effect  
`mf_shadow(...)`

⑤ Create a custom inset  
`mf_inset_on(x = mtq, pos = "bottomleft")  
mf_map(...)  
mf_inset_off()`

⑥ Create a world inset  
`mf_inset_on(x = "worldmap", pos = "right")  
mf_worldmap(mtg)  
mf_inset_off()`

⑦ Plot title  
`mf_title("A beautiful beach")`

⑧ Plot labels  
`mf_label(...)`

⑨ Plot annotation (in specific locations)  
`mf_annotation(...)`

⑩ North arrow  
`mf_arrow(...)`

⑪ Scale (in km)  
`mf_scale(...)`

⑫ Credits  
`mf_credits(...)`

## Export Maps

`mf_export()` exports maps in PNG or SVG formats.

The exported map width/height ratio will match the one of a spatial object.

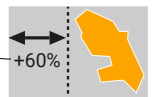
Additionally, `mf_export()` can be used to set a theme, to extend the map space on one or several side of the figure, or to center a map on a specific area.

Simple export (PNG)  
`mf_export(x = mtq, width = 500,  
filename = "my_export.png")  
mf_map(x = mtq, add = TRUE)  
dev.off()`

Export with a theme (SVG)  
`mf_export(x = mtq, width = 5, export = "svg",  
filename = "my_export.svg",  
theme = "nevermind")  
mf_map(x = mtq, add = TRUE)  
dev.off()`

Extra space on the figure (bottom, left, top, right)

`mf_export(x = mtq, width = 500,  
filename = "my_export.png",  
expandBB = c(0,0.6,0,0))  
mf_map(x = mtq, add = TRUE)  
dev.off()`



Export a map centered on a specific area  
`mf_export(x = mtq[30, ], height = 600,  
filename = "my_export.png")  
mf_map(x = mtq, add = TRUE)  
dev.off()`

## Further documentation

Vignettes on mapsf website: [riatelab.github.io/mapsf](https://riatelab.github.io/mapsf)

- > Get started
- > How to Use Themes
- > How to Export Maps
- > How to Create Inset Maps
- > How to Create Faceted Maps
- > How to Use a Custom Font Family